

Day 51 - Dynamic ARP Inspection (DAI)

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8:37 PM

DAI Info

ARP Review

ARP is a protocol that is used to learn the MAC address of another device with a known IP address.

- Typically, it is a two message exchange consisting of:
 1. **ARP Request**
 2. **ARP Reply**
- The **ARP** message itself is a Layer 2 message, but within the **ARP** header, it contains the **sender** MAC and IP, as well as the **target** MAC and IP.

Gratuitous ARP

A **Gratuitous ARP** message is an **ARP Reply** that is sent without receiving an **ARP Request**.

- It is sent to the **broadcast** MAC address.
- It allows other devices to learn the MAC address of the sending device without having to send **ARP Requests**.
- Some devices automatically send **GARP** messages when an interface is enabled, IP address is changed, MAC address is changed, and etc.

Dynamic ARP Inspection

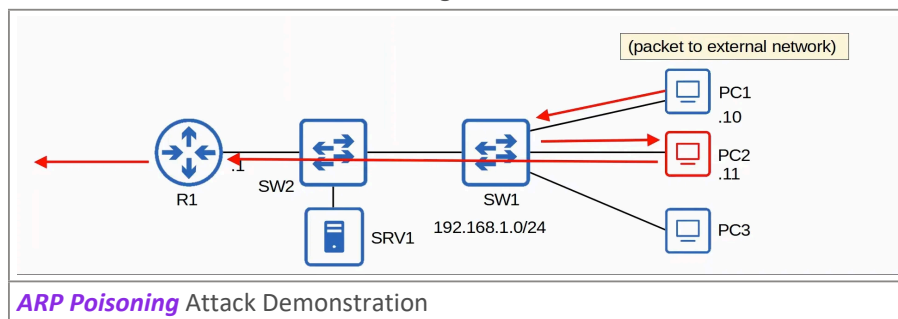
Dynamic ARP Inspection (DAI) is a security feature of switches that is used to filter ARP messages received on **untrusted** ports.

- DAI** only filters **ARP** messages. Non-**ARP** messages are not affected by it.
- By default, all ports are **untrusted**. There's also a standard to follow while assigning **trusted** ports.
 - Trusted** ports should be all the ports connected to other network devices (switches, routers)
 - Untrusted** ports should be all ports connected to end hosts.

ARP Poisoning (man-in-the-middle attack)

ARP Poisoning is an attack that's similar to the **DHCP Poisoning** attack. It involves an attacker manipulating **targets'** ARP tables so traffic is sent to the attacker.

- In order for the **attacker** to accomplish this, the **attacker** can send **GARP** messages using another device's IP address.
- Other devices in the network will receive the **GARP** and update their **ARP** tables, which will cause them to send traffic to the **attacker** instead of the legitimate destination.



DAI Operations

- When an **ARP** packet is received, **DAI** will do the following checks:
 - If a message arrived on a **trusted** port:
 - DAI** does not inspect the messages and they are forwarded as normal.

- If a message arrived on an *untrusted* port, do the following:
 - Examine the *sender* MAC and *sender* IP fields of **ARP** messages received on *untrusted* port, and checks that there is a matching entry in the **DHCP snooping binding table**.
 - Match = *Forward* the **ARP** message
 - No match = *Discard* the **ARP** message
- **ARP ACLs** can be manually configured to map IP/MAC addresses for **DAI** to check.
 - They are mainly used for hosts which do not utilize **DHCP**, therefore, they do not have an entry in the **DHCP snooping binding table**.
- **DAI** can also be configured to perform more *optional* in-depth checks.
- Like **DHCP snooping**, **DAI** also supports *rate-limiting* to prevent **attackers** from overwhelming the switch with **ARP** messages.
 - The reason this exists is because features like **DHCP snooping** and **DAI** both require work from the switch's CPU.
 - Even if an **attacker's** messages are blocked, they can overload the switch's CPU with excessive **ARP** messages.

DAI Configurations

- *show ip arp inspection*
 - Shows a summary of the **Dynamic ARP Inspection** configuration as well as statistics.
- *show ip arp inspection interfaces*
 - Shows the status of **Dynamic ARP Inspection** on the switch's interfaces.
- *ip arp inspection vlan VLAN-ID*
 - Enables **Dynamic ARP Inspection** on the specified VLAN.
 - (Unlike **DHCP snooping**, **DAI** doesn't have to be enabled globally beforehand)
- *ip arp inspection validate VALIDATION-MODE (SRC-MAC | DST-MAC | IP) ^arp-inspection-validate-cmd*
 - Configures the current **ARP Inspection** validation method(s).
 - Supports 3 validation modes, being *src-mac*, *dst-mac*, *ip*.
 - (You can use multiple by specifying them in the same command)
- *ip arp inspection trust*
 - Configures the current interface as trusted for **ARP** inspections,
- *ip arp inspection limit rate PACKETS burst interval SECONDS*
 - Limits how many **ARP** packets are allowed per *burst* interval
 - (Default settings are: **15** packets per **1** burst interval, in other words, **15** packets per second)
- *arp access-list ARP-ACL-ID*
 - Creates an **ARP ACL** and enters **ARP ACL** configuration mode.
 - In configuration mode:
 - { *permit* | *deny* } ip host *ip-address* mac host *mac-address*
- *ip arp inspection filter ARP-ACL-ID vlan VLAN-ID*
 - Applies an **ARP ACL** on the specified VLAN

DAI Rate-Limiting

DAI *rate-limiting* is enabled on *untrusted* ports by default with a rate of **15** packets per second.

- It is disabled on *trusted* ports by default.
- **DAI** also supports a *burst* interval, which means it can handle data that arrives in quick short bursts, and will not err-disable the interface (If the bursts don't go over the configured limit set through *ip arp inspection limit rate PACKETS burst interval SECONDS*).
- Can be re-enabled with *shutdown* ⇒ *no shutdown*, or through ErrDisable Recovery (*errdisable recovery cause arp-inspection*).

Difference between *rate-limiting* on DHCP snooping and DAI

1. DHCP snooping

- *Rate-limiting* is disabled on all interfaces by default.
- *Rate-limit* format: ==X packets per second==.
 - Default: **15** packets per second

1. DAI

- *Rate-limiting* is enabled on *untrusted* ports and disabled on *trusted* ports by default.
- *Rate-limit* format: ==X packets per Y seconds==.
 - Default: **15** packets per **1** second

DAI Optional Checks

There are optional DAI checks that you can enable (with [*ip arp inspection validate MODE*](#)) to further secure your network from attacks.

They include:

1. *src-mac*

- Enables validation of the *source* MAC address in the Ethernet header against the *sender* MAC address in the **ARP** body for *ARP Requests* and *ARP Responses*.
- The device classifies packets with different MAC addresses as invalid and drops them.

2. *dst-mac*

- Enables validation of the *destination* MAC address in the Ethernet header against the *target* MAC address in the **ARP** body for *ARP Responses*.
- The device classifies packets with different MAC addresses as invalid and drops them.

3. *ip*

- Enables validation of the **ARP** body for invalid and unexpected IP addresses. Addresses include (**0.0.0.0**, **255.255.255.255**), and all IP *multicast* addresses.
- The device checks the *sender* IP addresses in all *ARP Requests* and *ARP Responses* and checks the *target* IP addresses only in *ARP Responses*.

Optional checks and DAI

- These checks are done in addition to the standard **DAI** check (*sender* MAC/IP).
- If optional checks are configured, an **ARP** message must pass all of the checks to be considered valid.

DAI ARP ACL

For hosts which do not use DHCP, you can create an ARP ACL that will permit ARP messages from said devices.

- You'll need to create an **ARP ACL** with *arp access-list ARP-ACL-ID* and set a rule to permit **ARP** traffic from that specific host.
- Then you can apply it with **DAI** through the *ip arp inspection filter ARP-ACL-ID vlan VLAN-ID* command